

responsibilities are co-dependent and symbiotic. There will always be disagreement and misunderstandings, and occasional bad faith on both sides, but it is a relationship that has since Gutenberg at least served the Academy well as an essential engine of its activities. If I can conclude by mixing my metaphor, the diligence of academe may be under attack, but would it improve the situation if the passengers shot the coachman?

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Practical application of district compactness

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ABSTRACT. This is the fourth comparative study of compactness measures, as applied to political districting. We review the studies of Manninen, Young and Niemi and consolidate their listings into a single, more refined taxonomy that includes additional measures. We critique certain measures and conclude, with Niemi, that no single measure is perfect. While a few measures may be useful in comparing several redistricting plans for a single state, we argue that the prime value of quantifiable compactness is not in testing for gerrymandering, but as a criterion in a procedure designed to remove discretion from districting. From this perspective, a measure must be understandable and computable by the general public. The area/perimeter-squared measures are sound and workable when used with a pre-published catalogue of district 'building blocks', listing their areas and perimeters, and with data modified to eliminate excessive perimeter due to geography.

Introduction

While some students of the political districting issue in the USA defend the practice of political gerrymandering (Lowenstein and Steinberg, 1985), we hold it to be a violation of constitutional rights. We then face the question of what we would offer as an alternative to the discretionary districting that results in gerrymandering. We and others of like mind advocate districting procedures that require 'compactness', because we see that standard as an essential preventive to gerrymandering.

The original version of this paper (whose first unpublished draft circulated in 1987) was provoked by footnote 19 in Justice Stevens' concurring opinion in *Karcher v. Daggett* (462 US 756[1983])¹ and by Lowenstein and Steinberg's assertion that compactness is not a legitimate criterion for 'drafting or evaluating districting plans' (1985: 23). Young (1988) and Niemi *et al.* (1990) have covered some of the ground of our original version. While we have benefited from their analyses, we disagree with their public policy implications. Our interim research on districting plans drawn to satisfy objective criteria convinces us that there is more to say about compactness. This article² has three purposes:

1. To fill in a few gaps in Niemi *et al.*'s catalogue of compactness measures, revise their taxonomy somewhat, and build upon the critique of specific compactness measures begun by Manninen (1973) and continued by Young, and by Niemi *et al.*
2. To confirm Niemi *et al.*'s conclusion that 'the search for the measure of compactness is illusory'.
3. To make the case that, from a public policy perspective, the need is not for a measure—or combination of measures—that best 'proves' gerrymandering, but for a measure that works best in a procedure for drawing districts.

Lowenstein and Steinberg state 'compactness is neither simple nor straightforward' because 'there is no agreement on which method is best' (1985: 25). There is agreement, however, to be found on several important points. Those investigators who have made comparative studies all agree as to the unsoundness of certain measures. There is also agreement between ourselves and Niemi *et al.* that a combination of the Goedicke 'perimeter' measure and the Gibbs dispersion measure described below would be the most practical option if one were in a courtroom trying to convince a judge that a districting plan was gerrymandered. Lack of complete agreement may be attributed to insufficient effort expended in making rational comparisons of the various measures.

Geographers distinguish between two fundamental spatial concepts: shape and compactness. The concept of shape is explored in many writings, among which Taylor's (1971) is noteworthy. He demonstrates, among other things, that several parameters are necessary adequately to describe a given shape. Manninen (1973: 36–39) reviews Taylor's work and that of other geographers regarding shape and quotes Lee and Sallee (1970: 555): 'There exists no continuous one-to-one function from S , the set of all plane shapes, into R , the set of real numbers.' She concludes (1973: 40) that 'the development of a single number which can adequately express the two-dimensional quality of shape is impossible'.

Niemi *et al.* cite these statements (1990: 1159, note 6) about *shape* in support of their argument that no single measure of *compactness* adequately captures its multi-dimensional components of dispersion and perimeter. But Manninen (1973: 40), states that 'compactness... is but one aspect of the concept of shape'³ and that

Unlike the concept of shape, compactness can be adequately expressed as a single number. Attempts to measure compactness have been much more successful than attempts to retain all aspects of shape.

This confusion of these two concepts leads Niemi *et al.* to quote Lee and Sallee and Manninen's statements about apples (shape) in making their argument about oranges (compactness).

Taxonomy and critique of compactness measures

Manninen (1973: 56–75) analyzed 15 compactness measures, six of them mathematical transformations of a single formula. Young analyzed seven quantifiable⁴ compactness

measures, five of which had been dealt with by Manninen. His conclusions concerning the efficacy of the various measures agree with those of Manninen—with one exception. Niemi *et al.* produced a classification of 24 measures, including all the minor variations and mathematical transformations. Their listing included all seven of Young's measures and 10 of Manninen's 15. Our *Table 1* is a modification of Niemi *et al.*'s Table 1, expanded to include the remaining five of Manninen's measures and five others that we have added. Our listing follows their example in grouping together similar measures, but substantially refines the classification scheme and differs in some cases as to where measures belong.

Our critique of these measures will be brief where it concurs with earlier critiques, and longer where our assessment differs from earlier ones.

TABLE 1. A taxonomy of compactness measures

I. Shape-population measures	
A. District population compared with population of compact figure	
1.	Ratio of district population to population of minimum convex figure that completely contains the district (= Pop ₁). ⁵
2.	Ratio of district population to population of minimum circumscribing circle (= Pop ₂).
B. Other population measures	
1.	Population moment of inertia (Weaver and Hess) (= Pop ₃).
2.	Sum of all pairwise distances between centers of building blocks of district, weighted by building block populations (Papayanopoulos).
3.	Population moment about district population center divided by population moment about district geographic center (Iowa Reapportionment Statute).
II. Shape-only measures	
A. Area only	
1.	Absolute deviation from average area.
B. Perimeter only	
1.1	Sum of district perimeters (= Per ₁).
1.2	Sum of unutilized building block perimeters.
2.	Ratio of district perimeter to circumference of smallest possible circumscribing circle.
C. Dispersion measures	
1.1	Length-width difference
1.1.1	$L - W$, where L and W are measured on north-south and east-west axes, respectively (= Dis ₅).
1.1.2	$L - W$, where L is longest axis and W is maximum length perpendicular to longest axis (= Dis ₆).
1.2	Length-width ratio
1.2.1	W/L , where W and L are defined as in 1.1.2 (= Dis ₁).
1.2.2	W/L , where W and L are that of circumscribing rectangle with minimum perimeter (= Dis ₂).
1.2.3	W/L , where W and L are that of rectangle enclosing district and touching it on all four sides for which ratio of L to W is a maximum (= Dis ₃).
1.2.4	W/L , where L is the longest axis and W and L are that of rectangle enclosing district and touching it on all four sides (= Dis ₄).
2.	District area compared with area of compact figure
2.1	Ratio of district area to area of minimum circumscribing circle (Reock) (= Dis ₇).
2.2	Ratio of district area to area of minimum circumscribing regular hexagon (= Dis ₈).
2.3	Ratio of district area to area of minimum convex figure that completely contains district (= Dis ₉).
2.4	Ratio of district area to area of circle with diameter equal to district's longest axis (Gibbs) (= Dis ₁₀).

TABLE 1. (Cont.)

3. Miscellaneous

- 3.1 Average distance from district's areal center to the point on district perimeter reached by a set of equally spaced radial lines (Boyce and Clark) (= Dis_{12}).
- 3.2.1 Ratio of radius of largest inscribed circle to radius of smallest circumscribed circle (Haggett).
- 3.2.2 Ratio of radius of smallest circumscribed circle to 'minimum' radius.

D. Angular measure

1. Indentation Index: $(N - R)/(N + R)$, where N is the number of non-reflexive interior angles; R is number of reflexive interior angles (Taylor).

E. Area/perimeter quotients

1. Area/perimeter ratio (A/P).
2. Area/perimeter-squared ratio
 - 2.1 Area divided by perimeter-squared (A/P^2).
 - 2.2 Perimeter divided by square root of area ($P/\sqrt{A}$) (Grofman).
 - 2.3.1 $4\pi A/P^2$ (Cox) (= Per_2) (= Area of district divided by area of circle of equal perimeter [Miller]).
 - 2.3.2 $400\pi A/P^2$ (Goedicke) (Area of district as a percentage of the area of a circle of equal perimeter) (= CI).
 - 2.4 $1 - 2\sqrt{\pi A/P}$ (= Per_3).
 - 2.5 Perimeter of district divided by perimeter of circle of equal area ($P/(2\sqrt{\pi A})$) (Schwartzberg) (= Per_4).
 - 2.6 Perimeter of district as a percentage of minimum perimeter enclosing that area ($100P/(2\sqrt{\pi A})$) (= Per_5).

F. Relative moment of inertia

- 1.1 Second moment of district about its centroid divided by second (polar) moment of circle of area equal to that of district (Kaiser) (V).
- 1.2 $1/V$.
- 1.3 $1/\sqrt{V}$ (Blair and Biss) (= Dis_{11}).

*The parenthetical notations (Pop_1 , Per_1 , etc) appearing beside some definitions cross-index them to the listing of Niemi *et al.* (1990).

Criteria by which we judge which measure is 'best'

The *de facto* arbiter of what measure is best is intuition: which one most 'fully encompasses our intuitive notion' (Niemi *et al.* 1159), or which one best results in 'a correspondence between visual and quantitative expression' (Manninen: 75–76). While Young (1988: 106–107) demonstrates that visual intuition can be inconclusive in some cases, we can still discredit a measure by concocting two shapes one of which is obviously less compact than the other; and then showing that the measure, nevertheless, defines the former as being more compact.⁵ This is the technique employed by Young and us.

Analytical philosophers (Hospers, 1953: 51–53) have a term that applies to visual intuition in the case of compactness. They say that this term constitutes the *reportive* definition of compactness—the one most people have in mind when they use that word. Reportive definitions contrast with *stipulative* definitions in which one simply, and arbitrarily, stipulates that a given term will have such-and-such a meaning—regardless of whether that meaning is the one in the minds of most people. Table 1, therefore, gives 34 stipulative definitions of compactness: 34 different ways in which we can use numbers to quantify that term. The theoretically-best definition is the one that best conforms to the reportive definition, the one that, in all cases, gives numbers that rank various shapes in the order intuition says they should be ranked.

Shape–population measures

Niemi *et al.* (1990: 1165) note that all population measures stray from matching our visual 'reportive' definition. They demonstrate the unsoundness of measure I-A-2 with an example. Measure I-A-1 can be shown to be unsound in a similar manner. We might add that it is likely to prove exceedingly difficult, if not impossible, to determine the population of the 'minimum circumscribing circle', or 'other convex figure', because census counts are aggregated by blocks whose boundaries will not coincide with those of the circumscribing figure.

The remaining measures in this class, I-B-1, I-B-2 and I-B-3, are all unsound and complex. The first two of these measures depend not only on shape but size as well, thus favoring small urban districts over larger rural ones. The last one is described by the Iowa Legislative Service Bureau (1980). Despite being dimension-free, it has the following major problem: by this measure a perfectly compact district is one, regardless of shape, in which the population center coincides with the geographic center.

Shape-only measures

Area only; perimeter only. Both Manninen (1973: 65–66) and Niemi *et al.* (1990: 1160) reject arguments that size (Measure II-A-1) is relevant to compactness. We agree. The sum of district perimeters (Measure II-B-1.1) has appealed to various persons and groups (Tyler and Wells, 1971: 7; Adams, 1977: 874–875; Dixon, 1980–81: 847) and has been incorporated into a 1974 amendment to the Colorado constitution (1986, Article V, 47(1)). The *smaller* the sum, the more compact the *plan*. It is unsound because it is not dimension-free. Young (1988: 111–112) demonstrates this unsoundness, stating 'long perimeters do not necessarily imply noncompact districts' and 'use of the measure to compare alternative plans might tend to produce misshapen or gerrymandered urban districts at the expense of small improvements in the shape of rural districts whose boundaries are naturally much longer'.

The sum of unutilized building block perimeters (Measure II-B-1.2) has been employed by British Commonwealth geographers in their studies of constituencies drawn by the Parliamentary Boundary Commissions (Johnston and Rossiter, 1981a: 58; 1981b: 220–221; 1982: 137–138, 141 and 152; Rossiter and Johnston, 1981: 234) and in a study of hypothetical city commissioner districts in Mobile, Alabama (O'Loughlin and Taylor, 1982: 330–331). By this measure the most compact plan is that whose sum of unutilized building block perimeters is *largest*. Because this method is also not dimension-free, it could run into serious trouble where the polity to be districted is not homogeneous and has to be divided into districts, some of which contain areas of high population density and others, areas of low density. Its practitioners have recognized this problem (Johnston and Rossiter, 1981a: 65).

Measure II-B-2 is mentioned only in Justice Stevens's footnote 19 (462 US 756). He attributes it to Schwartzberg (1966), but Schwartzberg described no such measure. It can be easily demonstrated that this measure can give absurd results.

Dispersion measures: length versus width (II-C-1). Manninen (1973: 64–66), Young (1988: 109–110) and Niemi *et al.* (p. 1162, note 9) have explored length–width measures in all their nuances and have commented upon and/or demonstrated their unsoundness. We have nothing to add.

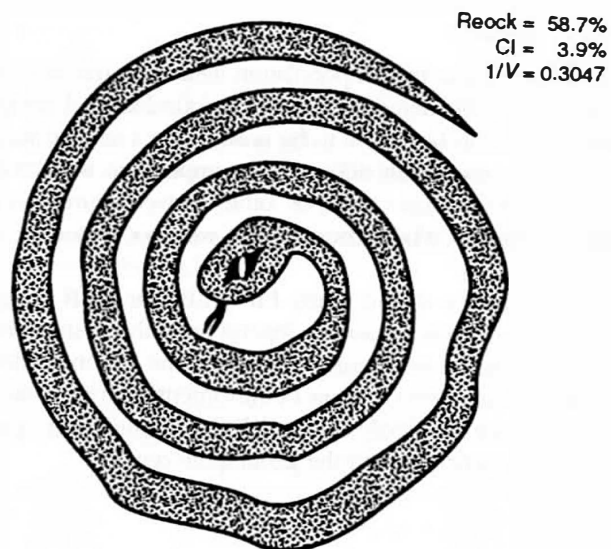


FIGURE 1. The coiled serpent (after Young, 1988).

Dispersion measures: district area compared with area of compact figure (II-C-2). Manninen (1973: 66–69) traces the evolution of compactness measures in this subclass, comparing Reock's (1961) measure (II-C-2.1) with that of Gibbs (1961) (II-C-2.4). Young (1988: 106–108) condemns the Reock measure, pointing out that, by it, the coiled serpent of Figure 1 is more compact than a square. Niemi *et al.* list two other measures in this subclass (II-C-2.2 and 2.3), calling them 'theoretically distinct'. They concur in Manninen's opinion (p. 69) that, of this subclass, Gibbs's 'longest axis circle' is the preferred option since, while being at no theoretical disadvantage, it is 'considerably easier to calculate'. Niemi *et al.* (1990: 1168–1176), Reock (1961), Manninen (1973: 92–112) and Backstrom *et al.* (1978: 1145, note 71, 1155) employed the Reock and Gibbs measures in empirical studies in several states, obtaining reasonable results. These measures are more unsound in theory than in practice.

Dispersion measures: miscellaneous. Measure II-C-3.1, involving radial distances from a district's center of gravity (Boyce and Clark, 1964) is arguably more of a shape measure than a compactness measure. It is found to be defective by Young (1988: 111) and by Niemi *et al.* (p. 1163, note 11). Its most serious fault is that a simple shape, such as a square, can have different indices depending on the selection of the direction of the first radial.

Measure II-C-3.2.1, proposed by Haggett (1966: 227–229), and measure II-C-2.3.2, described by Morrill (1981: 22), compare the radius of a figure's smallest circumscribed circle either to that of its largest inscribed circle or to its 'minimum' radius. Shapes can be concocted to show how either of these variants could yield counter-intuitive results. Morrill incorrectly attributes the 'minimum' radius version to Schwartzberg.⁶

Angular measure: indentation index (II-D). Taylor (1973) proposed a measure based on angles rather than on areas and distances, which he calls the Indentation Index I_A . Shortly after publication, Beth (Beth and Taylor, 1974: 1275) pointed out a defect in the measure, prompting Taylor to modify it (Beth and Taylor, 1974: 1276). Unaware of Beth's critique

and Taylor's modification, Young (1988: 110) drew figures demonstrating how the original formulation gave unreasonable results. Applying Taylor's modification does not invalidate Young's critique.

Area/perimeter quotients (II-E-1, II-E-2). The first area/perimeter formulation (II-E-1), A/P , differs from the remaining seven in not being dimension-free. The dimension problem of A/P can be remedied, however, by simply squaring the perimeter. This results in formulation II-E-2.1, which has been the compactness measure employed in all districting reform proposals introduced in the Ohio General Assembly since 1977.⁷ It gives a value of $1/4\pi$, or 0.07958, for a circle—which is not a convenient number to use as a standard of comparison.

The dimension problem of A/P can also be remedied by taking the square root of the area (measure II-E-2.2), which Grofman (1985: 85–86, note 4) did, resulting in another inconvenient value for a circle: 3.545.

The formulation of II-E-2.1 can be rendered more convenient by multiplying it by 4π , giving us formulation II-E-2.3.1. A circle then assumes the benchmark value of 1.0, a square of 0.7854, and less compact figures progressively lower values. Manninen traces the lineage of this formulation to Cox (1927). An identical result was obtained by Miller (1953) by the somewhat more laborious route of dividing the area of the shape by the area of a circle having the same perimeter.

A minor modification of II-E-2.3.1 suggested by Victor Goedicke, emeritus professor of mathematics at Ohio University, is to multiply it by 100, yielding the formulation II-E-2.3.2. We refer to $400\pi A/P^2$ as the Goedicke formula or as the Compactness Index (CI). Its values are percentages with a maximum of 100, the index of a circle.

Another variant of A/P^2 we list as II-E-2.4. Yet another variant is the one proposed by Schwartzberg (1966) (the measure he *did* propose), wherein the district's perimeter is compared to the perimeter of a circle of equal area. We list it as II-E-2.5. A final version we designate as II-E-2.6.

Niemi *et al.* classify these measures as 'perimeter measures' implying that they do little to measure dispersion. In fact, this subclass yields low indices whenever perimeter or dispersion is excessive, although excessive perimeter lowers a district's rating more than does excessive dispersion.

Manninen (1973: 61–63) states that this subclass of measures has 'one major weakness' in that they are 'too dependent on . . . perimeter'. She presents two shapes (Figures 2a and 2b). Although 2a and 2b have identical indices, most people would rate 2a as more compact. Young (1988: 108–109) independently arrives at the same conclusion in his evaluation of the Schwartzberg variant.⁸ To illustrate, he draws two shapes which we reproduce as Figures 2c and 2d. That 2d (CI = 32.0 percent) is, intuitively, more compact than 2c (solid outline; CI = 36.4 percent) is more of a judgment call. In order for 2c to attain a lower CI than 2d it must be elongated to the extent indicated by the dashed line. In both these examples, the discrepancy between intuition and what the numbers say is not great.

Relative moment of inertia measures (II-F-1)

Kaiser (1966) employed a relative second moment of area (measure II-F-1.1) to maximize compactness in his proposed districting procedure. To avoid the problem of dimension which beset the second moment measure employed by Weaver and Hess (1963), Kaiser divides the second moment of the district by the second moment of a circle having the

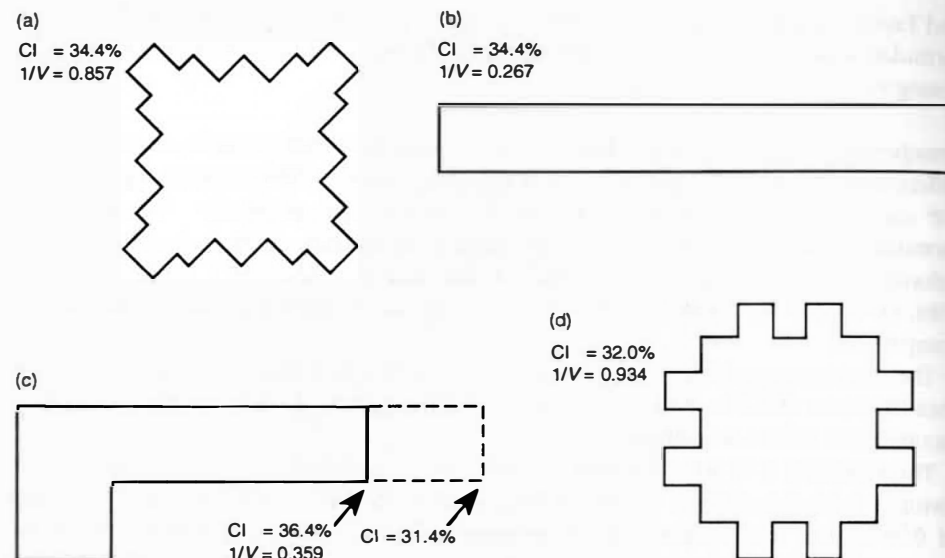


FIGURE 2. Perimeter measures ((a) and (b) after Manninen, 1973; (c) and (d) after Young, 1988).

same area as the district. Designating the *absolute* second moment of a district as T and its area as A , he defines a relative moment, V , of the district to be T divided by $A^2/2\pi$, the second moment of a circle of area A .

Because it is counter-intuitive for lower compactness values to be represented by higher numbers, we have listed the reciprocal of Kaiser's 'V'-values as measure II-F-1.2. Blair and Biss (1967: 6–10), with no reference to Kaiser, developed a measure that is identical to his except that it goes one step further and takes the square root of the reciprocal of 'V'. We list it as II-F-1.3. It is the only formulation of relative second moment cited by Niemi *et al.*⁹ and the only one evaluated by Manninen.

We find no merit in Blair and Biss's use of the square root of the reciprocal rather than simply the reciprocal. Not only does that operation add complexity, but it further compresses the range of values of a measure that already tends to exhibit a narrower 'spread' than many would prefer.

Manninen (1973: 112, 119) concluded that relative second moment is the best measure. On the other hand, Young (1988: 112) concluded that all measures are 'defective in some respect'. His consideration of moment of inertia tests was limited to the population moment of Weaver and Hess, which he correctly faulted for not being dimension-free. He did not consider the relative second moment measures of Kaiser (1966) and Blair and Biss (1967). He condemns the Weaver–Hess measure for giving 'good ratings to misshapen districts so long as they meander within a confined area', again using his 'serpent'. The serpent will definitely give a low population moment (good rating) to a district with a big city located in its head. However, if a relative second moment measure is applied to the serpent, the result will be a low compactness value. We did so and obtained a $1/V$ of 0.3047, and a Blair–Biss value of 0.552. These values would rank it below any of the districts in Morrill's plan for the Washington senate.

Niemi *et al.* follow Young into the same trap, stating that moment-of-inertia measures 'can be criticized for giving high scores to districts that "meander within a confined area"'

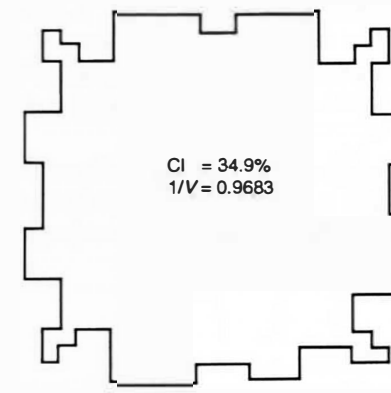


FIGURE 3.

(1990: 1163), thereby (mis)applying his critique of Reock's circumscribed circle and Weaver and Hess's population moment to their listing of the Blair and Biss relative second moment.

We have computed $1/V$ values for the Manninen and Young shapes used to question the soundness of area–perimeter measures and depicted in Figure 2. Note how these values are consistent with intuition. Although Young and Niemi *et al.* failed to discredit relative second moment measures by their analyses, we offer Figure 3 as evidence that even this class of measures can yield a counter-intuitive result. It has a $1/V$ value of 0.968, which is slightly higher than that of a square (0.955). We, therefore, affirm Niemi *et al.*'s conclusion that no *single* measure of compactness is theoretically perfect—although relative second moment comes close. We conclude, however, that the argument over whether there exists a theoretically perfect compactness measure is largely academic.

Application

Our major divergence from the three previous comparative studies concerns the purpose a quantitative measure of compactness might serve. Young and Niemi *et al.* assume it is for diagnosing partisan gerrymandering. Niemi *et al.* are most explicit, including in the title of their article 'the role of a compactness standard in a test for partisan gerrymandering'. We should examine what Niemi *et al.* learn from their empirical comparison of compactness measures, but first we should consider various ways in which compactness can be used to compare districting plans.

How compactness is used in comparing plans and their individual districts

Young states that a compactness measure 'should apply to the whole districting plan, as well as to individual districts' (1988: 112). That is not the problem. If a measure can tell us the compactness of an individual district, it is always possible to arrive at a value for the plan containing that district. But the reverse is not true: of the 34 measures we have considered, three cannot be used to compare the compactness of individual districts: the two sum of perimeter measures and the Weaver–Hess population moment.

1. *Sum, or average, of indices.* In general, the compactness of a plan would be the sum of the indices of its individual districts—or their average, which will always yield the same ranking. This approach, however, is flawed.

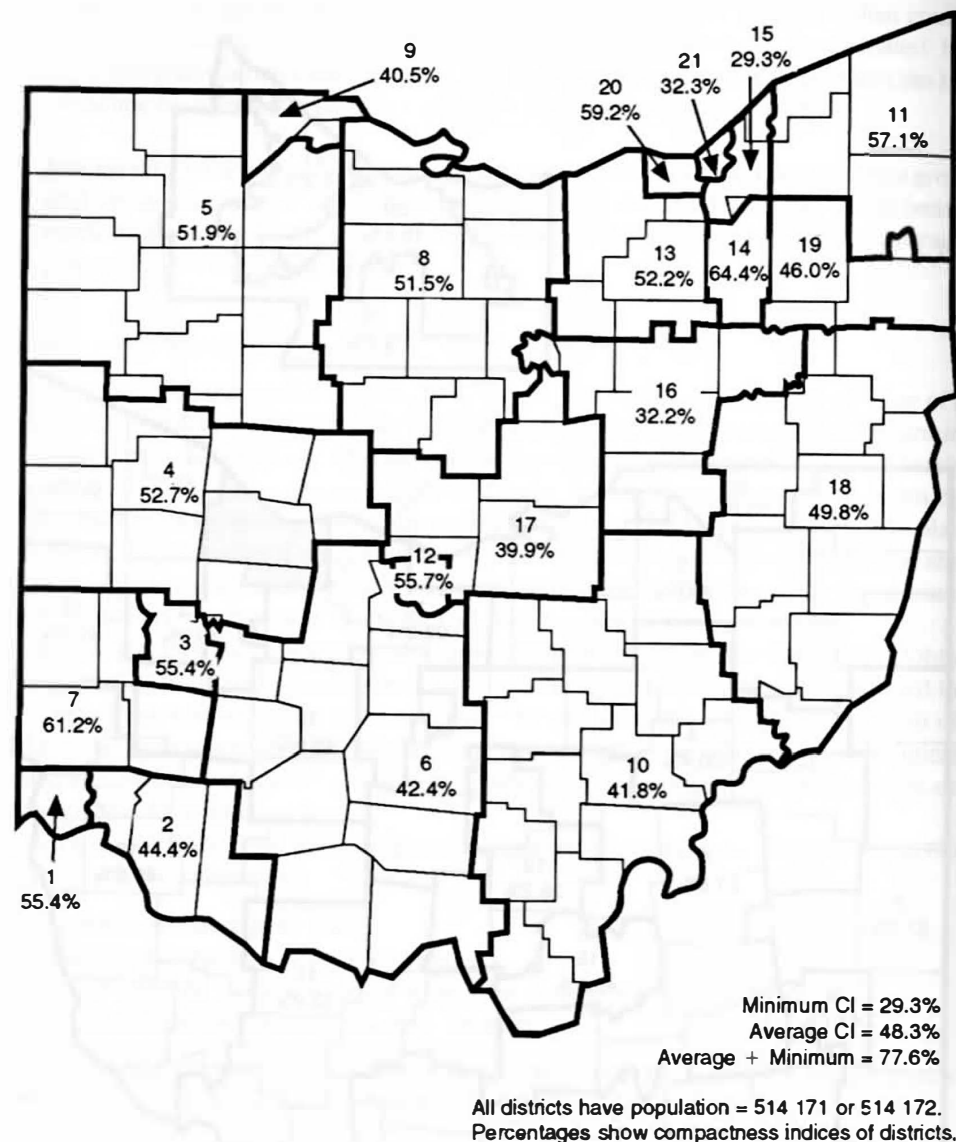


FIGURE 5. Ohio congressional districts, from the demonstration plan in the *amicus* brief submitted by the American Civil Liberties Union of Ohio, in *Flanagan v. Gillmor*.

Envision a courtroom scene in which plaintiffs are trying to nullify a plan on grounds of partisan gerrymandering and are offering various numerical measures of noncompactness as evidence. Should the US Supreme Court strike down a plan in such a situation, with no clear-cut remedy available, it could be making a serious mistake. If it made a judgment that, according to some test (one that, perhaps, included compactness measurements), a given plan had 'too much' bias—exceeding some arbitrarily-established cut-off point on a bias continuum—it might be stepping into a quagmire.

The dilemma posed by this quagmire, and the alternative of permitting gerrymandering to continue, can only be avoided if the courts assume a plan is unconstitutionally gerrymandered unless the authors (i.e. the state) can demonstrate that it was arrived at by an impartial process. Arguments supporting this view are made by Horn *et al.* (1989) and by Lawrence and Horn (1990). If they are valid, then what is needed is a carefully-designed procedure for drawing districts. Such a procedure will almost certainly include a quantifiable definition of compactness as one of its criteria. We thus arrive at the final purpose of this article: to determine what compactness measure works best in a districting procedure.

Compactness as a criterion in a districting procedure

Selection of measure. There is no sure-fire way—even with the most sophisticated computer technology—of determining what plan best satisfies a set of objective, quantifiable criteria when the database contains thousands of district 'building blocks'. Proponents of discretionless districting have, therefore, advocated putting out a database to the general public and inviting that public to use it to construct plans¹²—with the commitment to adopt whichever one best meets the criteria. This procedure is called competitive submissions. It has the added advantages, in our increasingly cynical and non-participatory electorate, of being open, of encouraging involvement, and of ensuring meticulous verification of the result.

While a compactness measure employed in such a procedure should be reasonably sound, and not be capable of producing grossly counter-intuitive results, it need not be theoretically perfect. However:

1. Of greatest importance is that it be adaptable to the use of what we call 'prepublished data' that will enable analytical—as opposed to graphical—computation of the index of any district in any plan.
2. Of almost equal importance is that individual plan-drawers be able to make those computations easily, with means they are certain to have (i.e. pocket calculators).
3. Finally, the measure should be as easy to comprehend as possible.

These requirements quickly narrow our choice of measures. Of the 34 we have considered, we can rule out as too unsound all but the four area-versus-area of compact figure measures, the seven area/perimeter-squared measures, and the three relative second moment measures. Of the first group, Gibbs's longest-axis circle is the easiest to compute; the other three measures do not capture any additional attributes of compactness that would justify their employment, despite their greater complexity. Of the third group, reciprocal of Kaiser ($1/V$) is best because its values go from 0 to 1.0 and it avoids Blair and Biss's square-root taking which undesirably compresses the range of values given. Of the second group, we should choose one that gives higher values for higher compactness and involves the least computation. That means we should use A/P^2 , $4\pi A/P^2$, or $400\pi A/P^2$. So we are now down to Gibbs, $1/V$, and three area/perimeter-squared measures.

Gibbs is less desirable than area/perimeter-squared for two reasons. First, it has a greater danger of yielding strongly counter-intuitive results, as Figure 1 demonstrates. The 'worst case' examples for A/P^2 are only mildly counter-intuitive, as Figure 2 demonstrates. Secondly, the length of the district's longest axis cannot be determined by an individual plan-drawer except by graphical measurement. Consider Figure 6.

POLITICAL DISTRICT BUILDING BLOCKS: MAHONING COUNTY									
Page 50-2									
City of Campbell (12, 577)	Population	Area km ²	Perimeter, km						
			a-b	b-c	c-d	d-e	e-f	f-g	g-h
Tract 8101	5,574	4.81	3.12	3.25	0.12	1.83	0.77	(f-a) 1.68	
Tract 8102	2,895	5.02	=f-a 8101 1.68	1.71	0.66	0.62	1.01	2.26	2.41
Tract 8103	2,230	1.45	=b-c 8102 1.71	0.98	(c-a) 0.79				
Tract 8104	1,878	0.92	=b-c 8103 0.98	=c-d 8102 0.66	1.03	(d-a) =d-e 8101 1.83			
City of Struthers (15, 343)									
Tract 8105	1,048	3.27	=c-d 8101 0.12	=c-d 8104 1.03	0.82	2.74	0.64	(g-a) 1.04	
Tract 8106	4,096	2.73	=c-d 8105 0.82	=e-f 8102 1.01	1.75	0.85	2.43	(f-a) 0.59	
Tract 8107	8,047	4.16	=e-f 8106 2.43	0.50	0.48	3.05	1.20	(f-a) 2.03	
Tract 8108	2,152	3.14	=f-a 8106 0.59	=f-a 8107 2.03	0.95	2.56	0.18	(f-a) =d-e 8105 2.74	
City of Lowellville									
Tract 8111	1,836	4.56		=e-f 8105 0.64	=e-f 8108 0.18	(d-a) 5.58			

FIGURE 6.

Figure 6 illustrates what we mean by 'prepublished data'. It shows a page from a hypothetical catalogue of district building blocks for Ohio to give the reader an idea of what the data would look like. With such data, the plan-drawer would, with certainty, know the CI of any district she draws because all she has to do is sum the building block perimeter segments that form the boundary of the district. However, she cannot, with these data, determine its longest axis. She must use a ruler.

Relative second moment, $(1/V)$, is theoretically superior to area/perimeter-squared in most cases. However, it would be simply impossible for an individual plan-drawer to know the indices of her districts prior to submission of her plan. If all the building blocks were completely described in some statewide co-ordinate system, the second moment of any district could be calculated, but it would be a vast and complex undertaking. The districting authority would be able to compute the indices of the districts as Manninen's

work again demonstrates. But the public would have to have complete trust in the people running the computer evaluating the plans. It could not even make its own rough, graphical calculations to keep the computer operators honest—as it could with the Gibbs measure.

The foregoing considerations lead us to a pragmatic selection of one of the area/perimeter-squared formulae. We earlier indicated a preference for Goedicke because it implies the idea of percentages, and the maximum value of 100 for a circle.

We cannot over-emphasize the importance of having pre-published data for district building blocks, such as illustrated in Figure 6, if districting is to be done by competitive submissions. Even if it were done by a legislature or by a commission, such a catalogue of building blocks would be necessary if the legislature or commission were to be subject to enforceable standards. The districting authority would most likely engage an engineering firm to make the necessary measurements to produce the database. It is the uniformity of the database that makes competitive submissions a standardized process in which no argument over the compactness of a district will arise that cannot be settled with a calculator.

Modification of data. Because of geography, in some states certain districts will inevitably have low compactness indices if the true boundaries of certain areas are used in the computations. Schwartzberg (1966: 445, 450) cites the coastal area of North Carolina as one example. Manninen deals with the Puget Sound area of Washington. In Ohio, Ottawa County with its peninsula and Lake Erie islands (Figure 7a) and Meigs County with its 'hook' at the big bend in the Ohio River (Figure 7b), are further examples. In a competitive submissions procedure, this state of affairs would not affect the ranking of plans because everyone would have low indices for the districts that included such areas and the losing plans would be those which, in addition to these unavoidable irregularities, had others due to manipulation of boundaries for political reasons.

While using true boundaries will not affect plan rankings in a competitive submissions procedure, their use does present problems if a compactness threshold (i.e. a minimum permissible value of compactness) is part of that procedure. Horn *et al.* (1989: 7) and Lawrence and Horn (1990: 33–34) have stated the rationale for such a threshold. If we use a compactness threshold, we want districts failing to meet it to do so because of political manipulation, not geography. Therefore, proponents have recommended two data modifications in the Ohio reform proposals. First, they specify that in calculating the index of any district, including the areas depicted in Figure 7, they be treated as if they were not

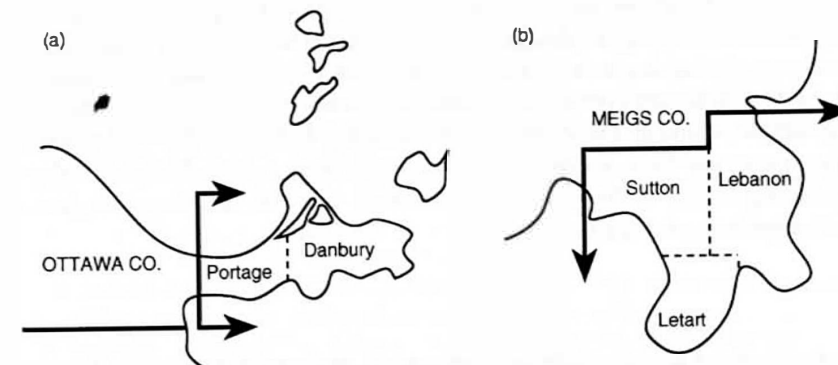


FIGURE 7.

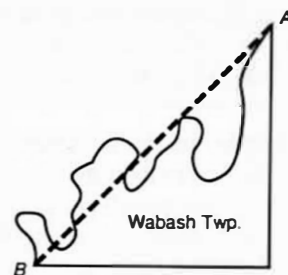


FIGURE 8.

part of the district, thus excluding both their areas and perimeters from the computation. Secondly, they recommend that in compiling the catalogue of district building blocks, those having irregular coastal or riverine boundaries be represented by straight-line segments, as appropriate. Figure 8 shows Wabash Township in Gibson County, Indiana. If the true perimeter of this building block is used along its north-west border, it contributes an excess of perimeter to any district containing it. Representing this perimeter by the line A-B solves this problem.

Dual thresholds. Empirical testing of the Ohio reform proposals over the past two years leads us to confirm the findings of Young (1988) and Niemi *et al.* (1990) that, even with the data modifications outlined above, there is a tendency for the area/perimeter measures to discriminate against urban districts, a certain amount of whose irregularity is due to the 'jagged edges' of census tract building blocks. In those states, a uniform threshold of, say, 30 percent tends to permit rural districts which, because they follow smooth county and township lines, are more elongated than we would like. Raising the threshold to penalize such districts, however, creates problems in the urban areas. A solution is to lower the threshold for districts in the big, urban counties (now 25 percent) and raise it in the remainder of the state (now 40 percent). This action illustrates the importance of empirical testing and experimentation to the practical application of district compactness.

Conclusion

We have reviewed the three previous comparative studies of compactness and concluded that no single measure always gives values that best agree with the reportive definition of compactness, though relative second moment comes close. Compactness measures are, ultimately, more likely to be of value in a districting procedure than as a means of proving gerrymandering during litigation. From the standpoint of a districting procedure, either the Cox or Goedicke area/perimeter-squared formulation is best. They have proven eminently simple, sound and workable in our empirical studies in Ohio and Indiana. It is very important to use these measures in conjunction with a pre-published catalogue of district building blocks if the districting procedure is to be unambiguous, free from subjective judgments, and fair to all.

Notes

- Footnote 19 cited 'a number of different mathematical measures of compactness' found in 'the scholarly literature'. It did not critique these measures or compare them.

- This is a condensation of our manuscript, District compactness: theory and application, copies of which are in several libraries and which may be located through the Online Computer Library Center (OCLC).
- Other aspects are elongation, orientation and centrality (Blair and Biss, 1967: 13-15).
- Young listed eight measures, the first (visual impression) of which was not quantifiable.
- In the case of a measure which is not dimension-free one can also discredit it by drawing a figure (or figures) of identical shape, but of different size or internal characteristics, and demonstrating that the measure gives different values.
- This marks the second attribution to Schwartzberg of a compactness measure he did not propose. The first was measure II-B-2.
- H.J.R. 90 (Locker), 112th General Assembly (G.A.), 1978; H.J.R. 15 (Locker) and S.J.R. 12 (Speck), 113th G.A., 1979; H.J.R. 8 (Locker) and S.J.R. 5 (Speck), 114th G.A., 1981; H.J.R. 35 (Lawrence) and H.J.R. 36 (Galbraith), 115th G.A., 1984; H.J.R. 33 (Lawrence) and S.J.R. 30 (Ney), 116th G.A., 1985; H.J.R. 7 (Lawrence), 117th G.A., 1987; H.J.R. 7 (Lawrence), 118th G.A., 1989; H.J.R. 4 (Lawrence), 119th G.A., 1991.
- In his listing of 'Desirable properties of compactness measures' Young also attacks the Schwartzberg measure by lumping it with Weaver and Hess as one that registers the size, as well as the shape, of the district thereby penalizing 'large rural districts'. This is not so. The Schwartzberg measure, like all other area/perimeter-squared measures, is dimension-free.
- It is listed in Niemi *et al.*'s Table 1 as 'Dis₁₁'. That listing contains two typographical errors and two incorrect citations. The formula that is given says 'Dis₁₀' instead of 'Dis₁₁' and omits the constant π that should precede the first integral sign. It attributes this measure to Schwartzberg—the third such incorrect attribution (see note 6)—and to Kaiser. Since the formula states the square root of the reciprocal of Kaiser's measure, the proper citation should be Blair and Biss.
- Flanagan v. Gillmor*, US Dist. Ct., S. Dist. of Ohio, Eastern Div. Case No. C-2-82-173, Brief of ACLU of Ohio as *Amicus curiae*, dated 17 October 1983 and 15 May 1985.
- In this case the state's plan was not a partisan one, but a bipartisan one in which critics say boundaries were manipulated to protect all Republican incumbents and all incumbent Democrats except for two it was designed to eliminate.
- These plans would most likely be manually drawn, but there would be no stopping anyone with a computer from using it to produce more trial-and-error solutions in less time. Since there would be no partisan advantage in winning such a competition, possession of a computer is no issue.

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Political conflict and foreign policy in the United States

A geographical interpretation

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ABSTRACT. Since the early 1970s, America's leaders have experienced great difficulty in mobilizing broad national support for their foreign policies. Deep and persistent conflict has been the norm, hampering national leaders' capacity to advance coherent and programmatic foreign policy. Most analysts identify ideological or institutional cleavages at the national level as the source of domestic political competition and conflict over foreign policy. This study offers an alternative, geographically-based interpretation. An analysis of voting in the Congress reveals that conflicts over the ends and means of American power have split the nation along regional lines, pitting the Northeast against the South. It is argued that these conflicts have revolved around the uneven costs and benefits of policies that were associated with the *Pax Americana* of the Cold War era. Like other periods in American history when there was little consensus over 'the national interest', today's conflicts are grounded in inter-regional struggles for political and economic advantage.

Introduction

During the quarter-century that followed World War II, American leaders were able to mobilize broad domestic support for their foreign policies. While the conventional wisdom that 'politics stopped at the water's edge' was at best a half-truth, the fact remains that political leaders enjoyed considerable latitude in the making of foreign policy. No one would characterize more recent American foreign policy-making in these terms. From the early 1970s onward, America's leaders experienced great difficulty in articulating a vision of the national interest that inspired broad support in Congress and the polity at large. The foreign policy consensus gave way to bitter and politically divisive conflicts over America's role in the world. Debates took on strongly emotional and symbolic overtones. The Cold War ended, but deep divisions over the ends and means of foreign policy persist. At a time when America's leaders need to make wise choices about the future, doubts remain about their ability to resolve the conflicts that have produced political gridlock and paralysis.

Practitioners, commentators and scholars recognize that American leaders no longer enjoy the freedom or autonomy in managing the nation's foreign policy that they once did. Disagreements arise when it comes to explaining this change. Some analysts locate the

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